Business Intelligence Project

Measure Definitions and Key Performance Indicators (KPIs)

1. Student and Enrollment Metrics

Total Students

Formula: DISTINCTCOUNT(dim_student[StudentSK])

Total Enrollments

Formula: COUNTROWS(fact_student_course)

Enrollments per Student

Formula: [Total Enrollments] / [Total Students]

Active Enrollments

Formula: COUNTROWS(fact_student_course where is_withdrawn = 0)

2. Performance Metrics

Average Score

Formula: AVERAGE(fact_student_assessment[score])

Pass Rate (%)

Formula: SUM(is_passed) / COUNTROWS(fact_student_assessment) × 100

Fail Rate (%)

Formula: 100 – Pass Rate

Weighted Average Score

Formula: SUMX(score × weight) / SUM(weight)

3. Progress and Outcome Metrics

Completion Rate (%)

Formula: SUM(is_completed) / COUNTROWS(fact_student_course) × 100

Withdrawal Rate (%)

Formula: SUM(is_withdrawn) / COUNTROWS(fact_student_course) × 100

Retention Rate (%)

Formula: 100 – Withdrawal Rate

4. Assessment Metrics

Submission Rate (%)

Formula: SUM(is_submitted) / COUNTROWS(fact_student_assessment) × 100

On-Time Submission Rate (%)

Formula: On-time submissions / Submitted assessments × 100

Average Days Late

Formula: AVERAGE(days_late where is_late = 1)

5. Demographic Metrics

Students by Gender (%)

Formula: Students per gender / Total Students × 100

Students by Region (%)

Formula: Students per region / Total Students × 100

Completion Rate by IMD Band (%)

Formula: Completion rate grouped by IMD band

6. Composite KPIs

Student Performance Index

Formula: 0.5×Score + 0.3×Submission Rate + 0.2×On-Time Rate

Module Success Score

Formula: (Pass Rate + Retention Rate + Completion Rate) / 3